ADAPTIVE GENOMIC VARIATION IN *RAPHANUS RAPHANISTRUM* IN URBAN ENVIRONMENTS

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BACKGROUND

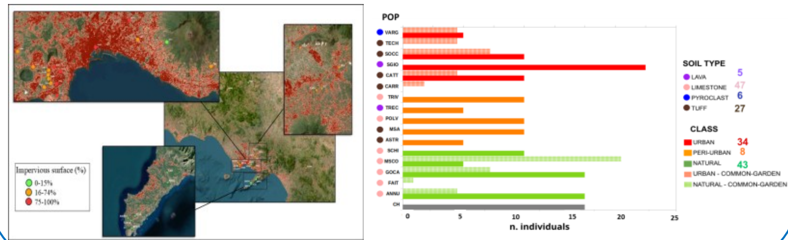
Raphanus raphanistrum (wild radish) is an edible annual plant of the family *Brassicaceae,* known for its competitive nature, rapid growth, and adaptability to diverse environments. Wild radish adaptation to urban environments and the underlying genomic mechanisms remain largely unexplored.

AIM

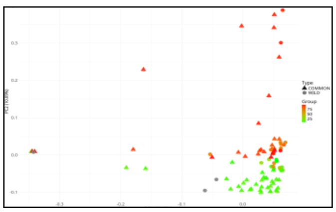
To investigate if and how genome-wide SNP variation drive adaptation to urban environments in *Raphanus.*

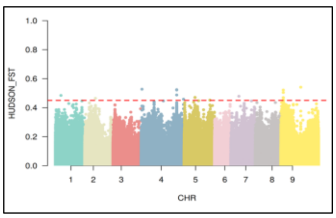
SAMPLING & METHODS

R. raphanistrum plants were collected from **16** sites across the province of Naples, representing a gradient of urbanization based on the percentage of impervious surfaces. Additionally, seeds were used in a common garden experiment; a total of **88** individuals, comprising **29** wild and **59** common-garden samples were collected for DNA extraction, variant calling, and SNPs identification.



POPULATION STRUCTURE



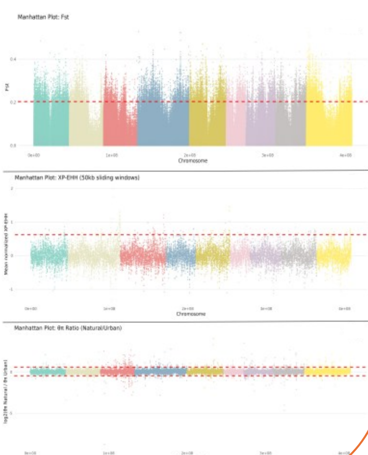


PCA analysis reveals distinct population structure between **URBAN** (red) and **NATURAL** (green) populations.

Several loci of interest were identified among SNPs exceeding the FST threshold of 0.45 within the coding or in upstream/downstream regulatory regions of genes involved in vascular development, transcriptional regulation, and stress response.

SELECTION SIGNALS

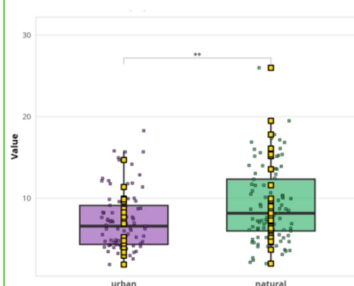
Overlap of XP-EHH, the θπ ratio, and F_{ST} identified several candidate SNPs. High-confidence candidates include Chromosome 2 variants in the *FLC* gene (flowering time regulator) and a *TTL3*-like gene (abiotic stress adaptation, 3 SNPs). Additionally, Chromosome 4 hosts a 6-SNP intergenic cluster between the *ING2* gene (epigenetic regulator of growth and stress) and an uncharacterized locus.



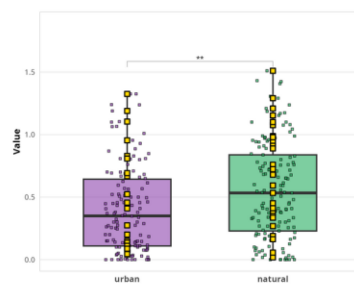
GWAS

URBAN plants exhibit reduced vegetative growth (leaf area) lower biomass (fruit weight) compared to **NATURAL** populations.

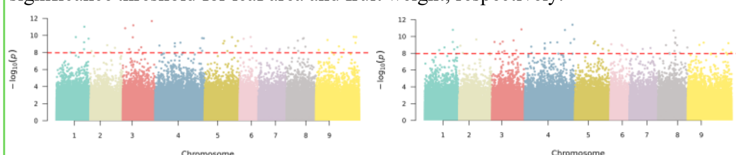
FRUIT WEIGHT



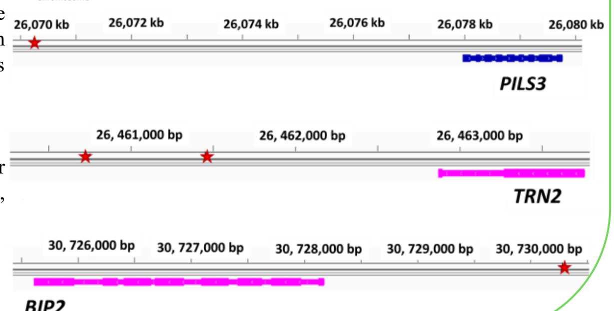
LEAF AREA



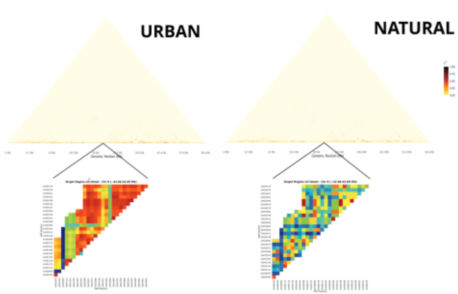
GWAS analysis identified **55 and 57 genomic regions** exceeding the Bonferroni significance threshold for leaf area and fruit weight, respectively.



4 loci are associated with identical SNPs showing concordant effect directions for both traits, indicating pleiotropic effects.



CHROMOSOME 9



Genome-wide LD analysis showed low r^2 values (~0.11) and rapid decay in both populations due to ongoing gene flow. However, a high-resolution scan of a critical region on Chromosome 9 (42.66–42.69 Mb, 27 SNPs) revealed distinct haplotype architectures. Strong LD ($r^2 > 0.8$) affected 41.38% of SNP pairs in **URBAN** populations versus only 6.21% in **NATURAL** ones. This localized, sharp increase in urban LD provides strong evidence of a recent selective sweep driven by urbanization.

OPEN QUESTIONS

- Do the genomic patterns identified in this study reflect global parallel evolution across diverse geographic urban gradients?
- What is the role of structural variants in adaptation to urban environments?
- Will functional validation experiments on the candidate genes confirm our results?